

The Trichotomous Agent

Body, Mind, and Soul in Economic Theory*

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Abstract

Standard economic models represent the agent as body (consumption, health) and mind (human capital, education). The soul does not appear. This paper introduces the *trichotomous agent*, an economic agent with three state variables: body (B), mind (M), and soul (S). The soul does not enter as another good to consume but *transforms the utility function itself*. The soul state reshapes time preferences, consumption behavior, social orientation, and moral constraints simultaneously through four channels; patience, satiation, generosity, and honesty. Grace, modeled as an exogenous stochastic process, provides the mechanism for spiritual transformation that the agent cannot self-generate. The model nests standard homo economicus as the special case where $S = 0$. We derive seven propositions connecting the soul state to corruption, savings, voluntary redistribution, total factor productivity, forgiveness equilibria, structural unemployment, and the existence of multiple steady states. Calibrated simulations show that a society moving from a low-soul to a high-soul equilibrium experiences a 60% increase in GDP, a two-thirds reduction in corruption, and a shift from revenge to forgiveness as the dominant social strategy. We argue that Christianity is uniquely positioned among world religions to supply the exogenous grace that the model requires, and that this claim is not merely theological but has testable economic implications. The model does not reject mainstream economics, but extends it by allowing the soul to be non-zero.

Keywords: spiritual capital, economic growth, utility transformation, grace, social capital, religion and economics, multiple equilibria

JEL Classification: O43, Z12, D91, O11

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1 Introduction

Why do some nations flourish while others, blessed with the same resources, the same institutions on paper, the same access to global knowledge, decay? The standard answers such as capital accumulation, human capital, technology, institutions explain much but leave a residual. That residual is large, persistent, and embarrassing. We call it total factor productivity and treat it as if we understand it, but we do not. We acknowledge a force we cannot fully measure or control, and we build entire frameworks around it.

This paper proposes that part of that residual has a name. It is the soul.

We introduce the *trichotomous agent*: an economic agent composed of body, mind, and soul. The body is physical health and vigor. The mind is human capital, e.g., knowledge, skills, cognitive capacity. The soul is spiritual capital that captures moral orientation, character, and relationship with God. The critical departure from standard economics is that the soul does not enter the model as another good to consume or another stock to accumulate. It *transforms the utility function itself*. A change in the soul state changes what the agent wants, how patient the agent is, how generous the agent is, and whether the agent can cheat. It is a meta-variable.

The intellectual ancestry of this idea is ancient. Augustine of Hippo, writing in the early fifth century as the Roman Empire crumbled around him, distinguished between two cities: the *civitas Dei* (City of God) and the *civitas terrena* (City of Man). The City of God is composed of those who love God above self. The City of Man is composed of those who love self above God. These two cities are intermixed in history, coexisting in every society, but they produce radically different outcomes such as different virtues, different institutions, and different destinies (Augustine, 426).

Augustine's two cities map precisely onto our model's two equilibria. *civitas Dei* is the high-soul equilibrium where patience, generosity, honesty, and satiation characterize the representative agent. The low-soul equilibrium is where impatience, greed, corruption, and insatiable desire dominate and is the economic expression of the *civitas terrena*. The critical insight, which Augustine understood and which our model formalizes, is that the transition between these two states cannot be accomplished by human effort alone. It requires grace.

1.1 What this paper does

We make five contributions:

1. We build a formal dynamic optimization model with three state variables (body, mind, soul) where the soul transforms preferences, not just the budget set. We derive the full Lagrangian, first-order conditions, and co-state equations.
2. We prove seven propositions connecting the soul state to observable economic outcomes: corruption (Proposition 1), savings and investment (Proposition 2), voluntary redistribution (Proposition 3), total factor productivity (Proposition 4), inter-group forgiveness (Proposition 5), structural unemployment (Proposition 6), and the existence of two stable equilibria (Proposition 7).

3. We show that the model *nests* standard homo economicus as the special case $S = 0$. The trichotomous agent extends mainstream economics; it does not reject it.
4. We present calibrated simulations showing that the quantitative effects of the soul state are large: a society moving from the low to the high equilibrium experiences GDP gains of approximately 60%, corruption falling by two-thirds, and a shift from revenge to forgiveness.
5. We argue on theological, philosophical, and empirical grounds that Christianity is uniquely suited among the world’s religions and ethical systems to supply the exogenous grace that the model requires. This is not a generic “religion is good for the economy” claim. It is a specific claim about the mechanism.

1.2 What this paper does not do

We do not claim that every Christian society has flourished or that every non-Christian society has failed. History is more complicated than any model. We do not claim that grace is the *only* determinant of economic outcomes. Physical capital, technology, geography, and institutions all matter, and our model includes them. We do not claim that the soul state is directly measurable with current data, though we discuss proxies. And we do not claim that this paper will convince the skeptic. We claim only that the framework is internally consistent, that it nests the standard model, that it generates testable predictions, and that it explains patterns that the standard model cannot such as persistent corruption, the failure of institutional reform, and the power of forgiveness.

1.3 Plan of the paper

Section 2 reviews the related literature and identifies the gaps our model fills. Section 3 argues for the distinctiveness of Christianity as the source of the model’s exogenous grace. Section 4 presents the formal model. Section 5 develops the four channels and seven propositions. Section 6 connects the model’s two equilibria to Augustine’s two cities. Section 7 presents numerical simulations. Section 8 proves the nesting result. Section 9 addresses the secular economist directly. Section 10 concludes. The full Lagrangian derivation and extended proofs are in the Appendix.

2 Related Literature

Our paper sits at the intersection of five literatures: the economics of religion, social capital, institutions and growth, culture and economic outcomes, and behavioral economics. In each case, we build on important findings while addressing a specific limitation.

2.1 Economics of religion

Iannaccone (1998) provides the foundational survey, establishing that religious participation is amenable to economic analysis: agents choose denominations, invest time in religious activities,

and respond to incentives within religious organizations. The economics of religion has since grown into a substantial field, treating religion as a club good, a form of insurance, or a source of network externalities.

Barro and McCleary (2003) take the field to macroeconomics. Using cross-country panel data, they find that religious *beliefs* (particularly belief in hell) positively predict economic growth, while church *attendance* has a negative effect conditional on beliefs, suggesting that the productive input is conviction, not time spent in pews. McCleary and Barro (2006) extend this analysis and propose a “religion market” framework where state regulation of religion affects both the supply and demand for spiritual goods.

Guiso et al. (2006) provide micro-level evidence that Christian denominations (particularly Protestantism) are associated with attitudes favorable to economic growth: trust, thrift, the belief that poverty results from laziness rather than injustice, and support for market competition.

What is missing. This literature establishes that religion *correlates* with economic outcomes but does not provide a formal model of the *mechanism*. Barro and McCleary treat religious beliefs as explanatory variables in a growth regression, but they do not model how beliefs enter the agent’s optimization problem. Guiso, Sapienza, and Zingales document attitudinal correlations but do not explain *why* Christian denominations produce different attitudes, or why those attitudes persist across generations. Iannaccone models the choice to participate in religion but not the effect of religious transformation on economic behavior.

Our contribution. We provide the missing mechanism. The soul state S enters the utility function and transforms preferences through four channels (patience, satiation, generosity, honesty). Grace ($\mathcal{G}_t$) is modeled as an exogenous stochastic process, or the formal analogue of what Barro and McCleary measure as “belief.” The model generates predictions that can be tested with the same data these authors use, but with a structural interpretation that their reduced-form regressions lack.

2.2 Social capital

Putnam (1993) demonstrated that civic engagement (measured by voter turnout, newspaper readership, and associational membership) predicts institutional quality across Italian regions, even after controlling for economic development. Putnam’s insight launched the social capital literature. Knack and Keefer (1997) extended it to the cross-country setting, showing that generalized trust (measured by the World Values Survey question “Can most people be trusted?”) predicts GDP growth after controlling for initial income, education, and investment rates. Fukuyama (1995) made the popular case that trust is the foundation of prosperity, arguing that “low-trust” societies (his examples: southern Italy, China, France) suffer from small firm size, weak institutions, and the costs of formal contracting.

What is missing. The social capital literature takes trust as a primitive; a cultural endowment that varies across societies but whose *origin* is left unexplained. Putnam traces Italian civic capital to medieval communes, but this pushes the question back: why did some regions develop communes and others did not? Knack and Keefer show that trust predicts growth but cannot explain why trust erodes in some periods and revives in others. Most critically, social capital is modeled as a *stock* that enters the production function or the constraint set, but it does

not *transform preferences*. An agent in a high-trust environment faces lower transaction costs (a constraint effect) but still maximizes the same utility function. If the environment changes, behavior reverts. This cannot explain the deep, persistent behavioral changes associated with genuine moral transformation.

Our contribution. We model the soul state as a *preference transformer*, not just a constraint shifter. Higher S does not merely reduce transaction costs. It changes what the agent wants, how patient the agent is, and whether the agent can cheat. This is a deeper form of social capital that persists even when the external environment deteriorates, because the change is internal. Furthermore, we provide an origin story for social capital: it derives from the soul state, which itself depends on grace and spiritual investment. Social capital, in our framework, is *endogenous to the soul*. It is the observable manifestation of an unobservable spiritual transformation.

2.3 Institutions and growth

North (1990) established the modern framework: institutions are the “rules of the game” (formal constraints like laws and constitutions, and informal constraints like norms and conventions) that structure economic exchange. Good institutions reduce transaction costs, protect property rights, and enable impersonal trade. Acemoglu and Robinson (2012) argue that the fundamental cause of differences in prosperity is *inclusive* versus *extractive* institutions, tracing institutional origins to historical contingencies (colonialism, the Black Death, the Glorious Revolution).

The institutions literature has been enormously productive. But it faces an increasingly uncomfortable empirical puzzle: *institutional reform does not reliably produce institutional quality*. Countries adopt constitutions, create anti-corruption agencies, establish independent judiciaries and central banks and yet corruption persists, rule of law decays, and growth stagnates. South Africa is the canonical example: one of the most admired constitutions in the world, robust Chapter 9 institutions, an independent Reserve Bank, and yet have governance indicators that have deteriorated relentlessly since 2008. The institutions-as-rules framework cannot explain why the same rules produce different outcomes in different societies.

What is missing. North himself acknowledged the gap. His informal constraints (norms, conventions, codes of conduct) do much of the explanatory work, but he provides no theory of where they come from or how they change. Acemoglu and Robinson’s historical contingency argument is powerful for explaining long-run divergence but offers no prescription for a society trapped in extractive equilibrium today. “Change your institutions” is advice that presupposes the existence of agents willing and able to design, implement, and sustain good institutions. Where do such agents come from?

Our contribution. We provide the missing micro-foundation for institutions. Good institutions require good agents; agents who are patient enough to invest in public goods, honest enough to resist corruption, generous enough to share power, and forgiving enough to sustain cooperation after conflict. These are precisely the characteristics that rise with the soul state S . The model predicts that institutional reform without spiritual transformation will fail (the institution operates in the low- S equilibrium and is captured by low- S agents), while spiritual transformation without institutional reform can still improve outcomes (high- S agents generate

trust, honesty, and cooperation even in weak institutional environments). This explains both South Africa’s institutional failure and Rwanda’s institutional success.

2.4 Culture and economic outcomes

Tabellini (2010) shows that cultural traits such as trust and respect for others, measured at the regional level in Europe predict economic development even after controlling for country fixed effects and historical institutions. He argues that culture is transmitted intergenerationally and co-evolves with institutions. Alesina and Giuliano (2015) survey the broader literature on culture and economics, documenting correlations between cultural variables (individualism, family ties, generalized morality) and outcomes (growth, innovation, redistribution). They emphasize that culture is slow-moving, persisting over centuries, and that it interacts with institutions in complex ways.

What is missing. The culture literature faces the same foundational problem as the social capital literature: culture is taken as given. Tabellini’s intergenerational transmission model explains persistence but not change. If culture is transmitted from parents to children and shaped by historical institutions, then a society born into a “bad” culture is trapped with no mechanism for cultural renewal. The literature documents correlations between cultural traits and economic outcomes but provides no formal model of how cultural traits enter the agent’s decision problem, no theory of cultural transformation, and no explanation for episodes of rapid cultural change (post-war Germany, post-genocide Rwanda, the East Asian tigers).

Our contribution. Our model provides exactly what the culture literature lacks: a formal mechanism for cultural change. The soul state S is the individual-level analogue of what the culture literature measures as “trust,” “respect,” or “generalized morality.” It is transmitted intergenerationally (through family devotional investment and the social multiplier ξ), it co-evolves with institutions (through the feedback between S and governance quality), and it is slow-moving in normal times (the low and high equilibria are locally stable). But, and this is the critical departure, it can change rapidly when grace intervenes. The exogenous grace shock ε_t^G is the model’s explanation for episodes of rapid cultural transformation that the intergenerational-transmission framework cannot account for. The Wesleyan revival, the post-war German *Stunde Null*, the Rwandan reconciliation are some examples of these grace shocks in our framework, and they produced exactly the S-curve transitions that the model predicts.

2.5 Behavioral and preference foundations

A growing body of work in behavioral economics has, independently and without invoking the soul, established micro-foundations for each of our four channels. These papers are important because they demonstrate that our model’s mechanisms are not ad hoc theological impositions but formalize behavioral regularities that mainstream economists have already documented.

Honesty. Mazar et al. (2008) develop a theory of “self-concept maintenance” in which individuals balance the financial benefits of cheating against the psychological cost of revising their moral self-image. Agents cheat only up to the point where dishonesty would force a negative update of their self-concept. This maps directly onto our honesty constraint $\theta_t \leq \bar{\theta}(1 - S_t)$: the soul state S determines the agent’s moral self-concept, which in turn sets the internal ceiling

on dishonesty. Mazar, Amir, and Ariely provide the behavioral evidence; we provide the state variable that moves the ceiling.

Identity and moral assets. Bénabou and Tirole (2011) formalize how agents invest in their identity because they care about “who they are.” In their model, past moral actions serve as signals of the agent’s type, and agents invest in moral behavior partly to maintain a favorable self-image that helps resist future temptation. This is the secular analogue of our soul as a state variable: identity is an accumulated asset that reshapes future behavior. The critical difference is that in their framework, identity investment is purely self-referential—the agent builds moral capital to serve the self. In ours, the soul is constituted by a relationship with God, and its deepest increments come from outside the agent entirely (grace).

Endogenous time preferences. Becker and Mulligan (1997) developed the foundational model showing how agents expend resources to “produce” patience, thereby raising their future discount factor β . Their framework demonstrates that differences in time preference can be endogenous to investment decisions, leading to divergent steady states in wealth and savings; a result that parallels our Proposition 2. Our patience channel $\beta(S)$ extends their insight by identifying a specific source of patience investment (devotional practice) and a specific mechanism for its transformation (grace), rather than leaving the “imagination technology” unspecified.

Warm-glow giving. Andreoni (1990) established that individuals derive internal, psychological utility from the act of giving itself—a “warm glow”—separate from the public good created. This is precisely what our generosity weight $\psi(S)$ captures: the soul state determines how much intrinsic satisfaction the agent derives from giving. Andreoni’s contribution is to show that warm-glow preferences rationalize observed charitable behavior that pure altruism cannot explain. Our contribution is to endogenize the warm glow: it is not a fixed taste parameter but a function of the soul state, which itself depends on grace.

What is missing. Each of these papers establishes one channel in isolation. None connects them. None asks why some agents have stronger self-concept maintenance, higher identity investment, greater patience, or warmer glow than others. And none provides a mechanism for *simultaneous* transformation across all four margins. Our model does: the soul state S is the single variable that moves honesty constraints, discount factors, generosity weights, and satiation parameters together. Grace is the shock that moves S . The behavioral literature has, unwittingly, been documenting the economic shadow of the soul, one channel at a time.

2.6 Summary: Positioning this paper

Table 1 summarizes how our model relates to each literature.

3 Why Christianity Is Distinct

Having situated our contribution in the economics literature, we now turn to the theological claim that distinguishes this paper from generic “religion and economics” work. Christianity is *qualitatively different* from other religious and ethical systems in ways that matter for the model’s mechanism. The distinction lies in three claims that are unique to Christianity, or at least uniquely central to it.

Table 1: Positioning the trichotomous agent model in the literature

Literature	Key insight we build on	What our model adds
Economics of religion	Religious beliefs predict growth (Barro & McCleary)	Formal mechanism: S transforms $\beta, \sigma, \psi, \theta$
Social capital	Trust predicts institutional quality (Putnam, Knack & Keefer)	Soul as <i>preference transformer</i> , not constraint shifter; origin of trust in grace
Institutions	Rules of the game matter (North, Acemoglu & Robinson)	Agents who sustain good institutions require high S ; institutions without soul are hollow
Culture & growth	Cultural traits persist and predict outcomes (Tabellini, Alesina & Giuliano)	Mechanism for rapid cultural change: exogenous grace shocks produce S-curve transitions
Behavioral & preferences	Each channel has independent micro-evidence (Mazar et al., Bénabou & Tirole, Becker & Mulligan, Andreoni)	Soul as the <i>single</i> state variable that moves all four channels simultaneously; grace as the mechanism

3.1 Grace is a gift, not an achievement

The central mechanism in our model is grace ($\mathcal{G}_t$): an exogenous shock to the soul state that the agent cannot manufacture, earn, or control. This is not a metaphor. It is a formalization of the Christian doctrine of *sola gratia*, or by grace alone.

In Christianity, salvation and moral transformation are gifts from God, received through faith, not achieved through human effort. “For it is by grace you have been saved, through faith—and this is not from yourselves, it is the gift of God—not by works, so that no one can boast” (Ephesians 2:8–9). The agent does not climb to God; God descends to the agent. This is the theological content of the exogeneity assumption.

Compare this with other major traditions:

- **Islam** emphasizes submission (*islam*) to God’s law and the performance of the Five Pillars. Salvation is earned through obedience; good deeds are weighed against bad deeds on the Day of Judgment. The moral transformation is endogenous since it depends on the agent’s effort and compliance. In our model, this would correspond to $\mathcal{G}_t = 0$ with a higher return on i_t^S (devotional investment), but no exogenous breakthrough.
- **Buddhism** teaches that liberation (*nirvana*) is achieved through the Eightfold Path; right view, right intention, right speech, and so on. The emphasis is on self-effort, mindfulness, and the extinction of desire. There is no external savior and no gift of grace. In our model, this corresponds to $\mathcal{G}_t = 0$ and a different objective function (minimizing desire rather than transforming it).
- **Hinduism** offers multiple paths (*yogas*) such as devotion, knowledge, action, and meditation, but in each case the emphasis is on the individual’s effort to achieve *moksha* (liberation from the cycle of rebirth). The *bhakti* (devotion) tradition comes closest to

the Christian concept of grace, but even here the devotee’s love is typically seen as the precondition, not the consequence, of divine favor.

- **Secular humanism** rejects the supernatural entirely and locates moral progress in reason, education, and institutional design. In our model, this corresponds to $S = 0$ (the soul is assumed away) and the entire burden of social improvement falls on the mind (M) and external institutions. As we will show, this is precisely the framework that has failed to explain persistent dysfunction in well-designed institutional environments.

The economic consequence of exogenous grace is profound. If moral transformation is endogenous, or if it depends entirely on the agent’s investment, then a society trapped in the low equilibrium *cannot escape by its own effort*. The shadow value of spiritual investment (μ^S) is low when S is low, the social multiplier works against the individual, and the rational response is to not invest. This is the poverty trap of the soul. Only an exogenous shock can break the cycle. Christianity claims that such a shock is available, always ($\bar{G} > 0$), and sometimes in extraordinary measure ($\varepsilon_t^G > 0$ which may include revival, awakening, and conversion).

3.2 Why not synergistic grace?

An alternative specification, natural to the Catholic and Arminian traditions, would model grace as *offered* universally but effective only when the agent cooperates through free will and devotional effort. Under such a formulation, $\mathcal{G}_t^{\text{eff}} = \mathcal{G}_t^{\text{offered}} \cdot h(d_t)$, where d_t is the agent’s devotional cooperation and $h(0) = 0$: if the agent refuses to cooperate, offered grace yields no transformation.

We reject this specification on three grounds: theological, economic, and empirical.

Theologically, synergistic grace contradicts the scriptural witness on the priority and sufficiency of divine action in salvation. Christ teaches that no one can come to Him unless the Father draws them (John 6:44). Paul insists that salvation “does not depend on human desire or effort, but on God’s mercy” (Romans 9:16) and that God “has saved us and called us to a holy life; not because of anything we have done but because of his own purpose and grace” (2 Timothy 1:9). The work of transformation, once begun, is God’s to complete: “He who began a good work in you will carry it on to completion until the day of Christ Jesus” (Philippians 1:6). These texts do not describe a cooperative project between God and the human will. They describe a sovereign rescue. The agent who has been genuinely transformed by effectual grace will indeed invest in devotion ($i^S > 0$), pray, worship, and serve. But these acts are *consequences* of regeneration, not conditions for it. To reverse the order, i.e., making human cooperation the trigger that activates divine grace, is to make salvation a joint venture and to relocate the decisive cause from God to the creature. This is precisely the error that the Reformation’s *sola gratia* was formulated to correct.

Economically, synergistic grace dissolves the paper’s most distinctive contribution. If grace is activated by human devotional effort, then *any* religion with a devotional practice can produce spiritual transformation. Islam’s Five Pillars, Buddhist meditation, Hindu *bhakti*, and Confucian ritual all involve substantial devotional investment ($d_t > 0$). The model would predict that all of these are equally capable of raising S and producing the high equilibrium. The paper’s specific

claim about Christianity (Section 3, Contribution 5) would collapse into a generic “religion is good for the economy” thesis—precisely the undifferentiated claim that we distinguish ourselves from in Section 2.

Moreover, synergistic grace eliminates the poverty trap of the soul. If agents can incrementally cooperate their way to higher S , the low equilibrium becomes escapable through collective moral effort. This sounds appealing but contradicts the central empirical puzzle the paper addresses: why do societies with excellent institutions (South Africa’s Constitution, Chapter 9 bodies, independent Reserve Bank) nevertheless stagnate in corruption and dysfunction? If incremental cooperation could bootstrap a society out of the trap, institutional reform should have worked. It did not.

Empirically, some examples often cited in favor of synergistic grace such as the secularization of Quebec in the Quiet Revolution, Ireland’s rapid de-Christianization, are already explained by our model without modifying the grace specification. Spiritual depreciation δ^S operates continuously. When a prosperous society reduces devotional investment ($i^S \rightarrow 0$), the soul decays toward the grace floor $\bar{G}/\delta^S$. The mechanism is standard capital depreciation, not a withdrawal of synergistic cooperation. Rising wages increase the opportunity cost of devotional time (this is already captured in the equal-marginal-returns condition, equation 12), and the rational agent substitutes toward mind and body investment. Secularization is endogenous to the *investment decision*, not to the *nature of grace*. The original model explains these patterns; the synergistic revision adds complexity without explanatory gain.

3.3 Transformation, not compliance

The second distinctive feature of Christianity, as formalized in our model, is that grace transforms *preferences*, not just *constraints*.

Most ethical and religious systems operate on the constraint set. “Do not steal” is a constraint. “Pay your taxes” is a constraint. “Give 2.5% of your wealth to charity” (the Islamic *zakat*) is a constraint. These rules can improve outcomes within a given utility function; they are external enforcement mechanisms that change the feasible set without changing what the agent wants.

Christianity does something different. It claims to change what the agent wants. “Therefore, if anyone is in Christ, the new creation has come: The old has gone, the new is here!” (2 Corinthians 5:17). The Christian claim is not merely that you should be generous, it is that you will *want* to be generous, because your heart has been changed. Not merely that you should be honest, but that dishonesty will become abhorrent to you. Not merely that you should forgive, but that forgiveness will flow naturally from the experience of having been forgiven.

In our model, this is captured by the soul’s transformation of the utility function through four channels: the discount factor $\beta(S)$ rises (patience), the curvature $\sigma(S)$ rises (satiation), the generosity weight $\psi(S)$ rises, and the honesty constraint $\theta \leq \bar{\theta}(1 - S)$ tightens. The agent does not merely face tighter rules, the agent becomes a *different kind of agent*. This is the formal content of sanctification.

3.4 Norms are derivative, not foundational

The economics literature on social capital treats norms (trust, cooperation, civic virtue) as foundational variables (Putnam, 1993; Knack and Keefer, 1997). We argue that norms are *derivative*. They are the visible expression of an underlying soul state, not the cause of it.

Consider the empirical puzzle: why do some cultures produce trust and others produce corruption? If norms are foundational, this question has no answer; norms are just *there*, transmitted by families and institutions, and if you are unlucky enough to be born into a low-trust culture, that is your fate. But if norms are derivative, if they flow from the soul state, which itself depends on grace and spiritual investment, then there is a causal structure beneath the surface, and the possibility of change.

The historical record supports this. The abolition of slavery, the development of hospitals, the creation of universities, the codification of human rights, the emergence of modern science all have deep roots in Christian thought and Christian communities (Stark, 2005; Woodberry, 2012). These are not coincidences. They are the institutional expression of a soul state shaped by grace.

The secular economist may respond: “Fine, but I can explain all of this with social capital and cultural evolution without invoking God.” We address this objection directly in Section 9. For now, we note only that the secular account cannot explain why social capital emerges in some places and not others, why it decays in some periods and revives in others, or why institutional reform so frequently fails to produce the behavioral change it was designed to elicit. Our model can.

3.5 Relationship to Weber

The most famous argument connecting Christianity to economic outcomes is Max Weber’s *Protestant Ethic and the Spirit of Capitalism* (Weber, 1905). Weber argued that Calvinist anxiety about predestination (the agonizing uncertainty over whether one was among the elect) drove Protestants to seek worldly “signs of election” through disciplined work, thrift, and capital accumulation. The unintended consequence was capitalism.

Our model differs from Weber in three fundamental respects.

First, the *mechanism* is different. Weber’s agent works hard because of *anxiety* (the fear of damnation, the desperate search for assurance). Our agent is transformed by *grace*; the experience of unmerited love that produces gratitude, not fear. Weber’s agent accumulates capital to reassure himself. Our agent invests because a transformed heart values stewardship, sees the needs of others, and thinks long-term. The psychological and theological foundations are opposite: anxiety versus peace, earning versus receiving, performance versus gratitude.

Second, the *scope* is broader. Weber focused on one channel: work ethic → capital accumulation. Our model identifies four simultaneous channels (patience, satiation, generosity, honesty) and derives seven propositions spanning corruption, savings, redistribution, TFP, forgiveness, unemployment, and multiple equilibria. The Weberian thesis is, in effect, a special case of our model restricted to the patience and satiation channels.

Third, the *applicability* is wider. Weber’s thesis was historically specific. It explained the rise of capitalism in 16th–17th century Northern Europe. Our model applies to any society

at any time: the soul state S and its transformation by grace are not confined to a particular denomination or era. The model can speak to 21st-century South Africa, post-genocide Rwanda, or post-war South Korea, contexts far removed from Weber’s Calvinist burghers.

We regard Weber as an important precursor, not a rival. His insight that theological commitments have economic consequences was correct. What he missed was the mechanism (grace, not anxiety) and the full scope of the transformation (preferences, not just effort).

3.6 The dark side: When Christianity fails

An honest paper must address the obvious objection: Christianity has also produced Crusades, Inquisitions, slave-trading nations, and directly relevant to this paper, the apartheid theology of the Dutch Reformed Church in South Africa.

Our model explains this, rather than being embarrassed by it. Nominal Christianity such as church membership, doctrinal assent, and cultural identification, is not the same as a high soul state. The model’s predictions depend on S , not on religious affiliation. An agent who attends church but has $S \approx 0$ (no genuine transformation, no operative grace) behaves exactly like *homo economicus*: self-interested, impatient, dishonest up to the constraint. The apartheid theologians who blessed racial oppression were, in our model’s terms, low- S agents who used the language of Christianity to sanctify the *libido dominandi*. Augustine would have recognized them immediately as citizens of the City of Man dressed in the garments of the City of God.

The prosperity gospel (the claim that God rewards faith with material wealth) is the contemporary version of this corruption. It inverts every channel in our model: it encourages *impatience* (demand your blessing now), *insatiability* (God wants you rich), *selfishness* (your faith, your reward), and *dishonesty* (fake miracles, exploitative tithing). In our framework, the prosperity gospel is not high- S Christianity. It is a $S \approx 0$ ideology that mimics the vocabulary of grace while denying its substance.

The testable prediction is clear: societies with high rates of *genuine* religious practice (measured by behavioral proxies such as volunteering, charitable giving, marital stability, honesty in experiments, rather than self-reported belief) should outperform societies with high rates of nominal affiliation. The emerging empirical literature on “believing versus belonging” supports this distinction (Barro and McCleary, 2003).

3.7 The corrosion of works-based religion

The prosperity gospel is the most visible corruption of Christianity, but there is a subtler and more widespread one: the belief that religious works *earn* spiritual standing. This is the works-based framework, and it does not merely fail to raise the soul state, but can actively corrode it.

Consider an alternative model in which charitable giving g_t enters the soul’s law of motion: $S_{t+1} = (1 - \delta^S)S_t + \phi^S(i_t^S) + \kappa g_t + \mathcal{G}_t$, where $\kappa > 0$ rewards giving with spiritual capital. In such a framework, the first-order condition for giving changes. The standard model yields $\psi(S_t)g_t^{-\sigma} = \lambda_t$: the agent gives because the soul makes giving intrinsically satisfying. The works-based model adds an investment return: the agent also gives because giving raises S ,

which raises β (more patience), raises σ (more satiation), and relaxes future constraints. Giving becomes an *investment with private returns*, not an expression of genuine other-regard.

This has three destructive consequences.

First, it contaminates the generosity prediction. In our model, $g = 0$ when $S = 0$ (Proposition 3): a soulless agent has no reason to give. In the works-based model, $g > 0$ even at $S \approx 0$, because the agent gives instrumentally to build spiritual capital and harvest the patience and satiation dividends. The model can no longer distinguish genuine generosity from strategic philanthropy. The Pharisee who gives ostentatiously to be seen (Matthew 6:2) and the widow who gives her last mite from a transformed heart (Mark 12:43–44) produce the same observable g , but the Pharisee’s soul is not growing. It is calcifying.

Second, works-based religion produces spiritual pride, or the conviction that “I have earned my standing before God through my moral effort.” Pride is the quintessential low- S disposition. Augustine identified it as the root of the *libido dominandi*: the desire to dominate, to be above others, to be self-sufficient. In the language of our model, spiritual pride *lowers* ψ (the proud agent serves the self, not others), *lowers* σ (spiritual achievements become a form of insatiable consumption. The Pharisee always needs more visible righteousness), and *loosens* the honesty constraint (the self-righteous agent can justify dishonesty toward those deemed less worthy). The agent who believes she has earned her soul state is, paradoxically, further from genuine transformation than the agent who knows she has not. “God opposes the proud but shows favor to the humble” (James 4:6).

Third, works-based religion unravels the multiple equilibria result (Proposition 7). If g feeds back into S , the soul equation gains an additional convexity: agents at moderate S give, which raises S , which raises giving further. The unstable tipping point $\bar{S}^*$ softens into a smooth gradient, and the sharp “critical mass” threshold dissolves. The model loses its ability to explain why societies cluster at two distinct poles, high-trust flourishing or low-trust dysfunction, rather than distributing continuously along a spectrum. The empirical pattern is bimodal, not smooth (Knack and Keefer, 1997; Putnam, 1993). The works-based model cannot generate this.

The deeper point is theological. Grace transforms; works perform. The agent transformed by grace gives because giving has become natural to her; the utility function has changed. The agent performing works gives because giving is profitable to her; the utility function is unchanged, but the investment set has expanded. The first is sanctification. The second is sophisticated self-interest dressed in religious clothing. And the historical record, from the Pharisees to the medieval indulgence trade to the modern prosperity gospel, suggests that the second does not merely fail to produce the high equilibrium. It actively entrenches the low one, by giving agents the *appearance* of high S (visible piety, conspicuous charity) while preserving the *substance* of low S (pride, insatiability, instrumentalism). Christ’s harshest words were reserved not for the openly sinful but for the religiously self-righteous (Matthew 23). The model should reflect this: works do not build the soul. Grace does.

4 The Model

Consider an agent who lives for T periods and makes decisions about consumption, education, and devotion. The agent has three state variables:

- $B_t \in [0, 1]$: **Body**—physical health and vigor.
- $M_t \in [0, \infty)$: **Mind**—human capital (knowledge, skills, cognitive capacity).
- $S_t \in [0, 1]$: **Soul**—spiritual capital (moral orientation, character, relationship with God).

4.1 Preferences

The agent maximizes lifetime utility:

$$V_0 = \mathbb{E}_0 \sum_{t=0}^T \beta(S_t)^t u(c_t, g_t; S_t) \quad (1)$$

where c_t is own consumption, g_t is giving (voluntary transfers to others), and S_t is the soul state. The soul enters the utility function in two ways:

1. It transforms the **discount factor**: $\beta(S_t) = \underline{\beta} + (\bar{\beta} - \underline{\beta}) S_t$, where $0 < \underline{\beta} < \bar{\beta} < 1$.
2. It transforms the **shape** of the period utility function.

The period utility function is:

$$u(c_t, g_t; S_t) = \frac{c_t^{1-\sigma(S_t)}}{1-\sigma(S_t)} + \psi(S_t) \frac{g_t^{1-\sigma(S_t)}}{1-\sigma(S_t)} \quad (2)$$

where:

- $\sigma(S_t) = \underline{\sigma} + (\bar{\sigma} - \underline{\sigma}) S_t$ is the curvature parameter (inverse elasticity of intertemporal substitution). Higher S means higher curvature meaning the agent is more easily satisfied with material goods.
- $\psi(S_t) = \psi_0 S_t^\eta$ with $\eta > 0$ is the weight on giving. When $S = 0$, giving yields no utility and the agent is purely self-interested.

4.2 Budget constraint

The agent earns legitimate income from human capital applied to production. In addition, the agent has access to a pool of appropriable rents $\Pi_t > 0$ (public resources, positions of trust, contractual opportunities where cheating is feasible) and can divert a fraction θ_t of these rents to private use:

$$c_t + g_t + i_t^B + i_t^M + i_t^S = w_t M_t B_t + \theta_t \Pi_t \quad (3)$$

where w_t is the wage per effective unit of labor, $M_t B_t$ is effective labor (human capital weighted by health), and i_t^B, i_t^M, i_t^S are investments in body, mind, and soul respectively.

The rent pool Π_t represents resources that are vulnerable to appropriation: government budgets, aid flows, tax revenues, contractual obligations, employer assets, communal property. In a developing economy with weak institutions, Π_t is large. The critical feature is that corruption *benefits the individual* ($\theta_t \Pi_t$ enters the agent's income) while *harming aggregate output* (resources diverted from productive use reduce effective labor; see Section 5). This wedge between private gain and social cost is the fundamental tension that the honesty constraint must resolve.

Remark (Where do rents come from?). We treat Π_t as exogenous to the individual agent, determined by the institutional environment. In a richer model, Π_t could be endogenized: weaker governance (itself a function of $\bar{S}$) creates larger rent pools, generating a feedback loop where low soul states produce weak institutions that produce more temptation. For the present analysis, taking Π_t as given is sufficient to establish the mechanism.

4.3 The honesty constraint

The agent faces an internal moral constraint on dishonesty:

$$\theta_t \leq \bar{\theta} (1 - S_t) \quad (4)$$

where $\bar{\theta} \in (0, 1)$ is the maximum feasible appropriation rate. When $S_t = 1$ (fully Christ-centered), the constraint forces $\theta_t = 0$ which implies that the agent *cannot* cheat because internal moral conviction prevents it. When $S_t = 0$, the constraint is slack and the agent appropriates up to $\bar{\theta}$ of available rents.

This is not an external enforcement mechanism. There are no police, no courts, no regulators in this constraint. It is the formalization of conscience; an internal limit that tightens as the soul grows. The agent *wants* to cheat (it raises income by $\theta_t \Pi_t$) but the soul prevents it.

4.4 Laws of motion

Body. Physical health depreciates with age and can be maintained through health investment:

$$B_{t+1} = (1 - \delta_t^B) B_t + \phi^B(i_t^B) \quad (5)$$

where δ_t^B increases with t (aging) and $\phi^B(\cdot)$ is concave. The terminal condition is $B_T = 0$ (death).

Mind. Human capital accumulates following Ben-Porath (1967) with depreciation:

$$M_{t+1} = (1 - \delta^M) M_t + \phi^M(i_t^M, M_t) \quad (6)$$

where $\phi^M(i_t^M, M_t) = A^M (i_t^M)^\gamma M_t^{1-\gamma}$ is the Ben-Porath production function. Existing human capital aids the acquisition of new human capital.

Soul. The soul evolves according to:

$$S_{t+1} = (1 - \delta^S) S_t + \phi^S(i_t^S) + \xi \bar{S}_t + \mathcal{G}_t \quad (7)$$

where δ^S is the spiritual depreciation rate (decay from neglect), $\phi^S(i_t^S)$ is a concave production function for spiritual growth through devotional investment (prayer, worship, fellowship, study of Scripture), $\xi \bar{S}_t$ is a social multiplier capturing peer effects from the average soul state $\bar{S}_t$ in the community (it is easier to grow spiritually when surrounded by others doing the same), and $\mathcal{G}_t$ is grace. The parameter $\xi > 0$ governs the strength of this complementarity and is the key source of multiple equilibria (Proposition 7).

4.5 Grace

Grace follows:

$$\mathcal{G}_t = \bar{G} + \varepsilon_t^G, \quad \varepsilon_t^G \sim F(\cdot), \quad \varepsilon_t^G \geq 0 \quad (8)$$

The two components correspond to a theological distinction. $\bar{G} > 0$ is *common grace*: God’s persistent kindness toward all creation such as sustaining life, restraining evil, granting the rain to fall on the just and the unjust (Matthew 5:45). Common grace is universal and never withdrawn. It is the reason that even thoroughly secular societies retain a residual moral sense, that cooperation does not collapse entirely, and that beauty persists in a fallen world. The stochastic term ε_t^G represents *effectual grace*: the specific, transformative work of the Holy Spirit that regenerates the heart and grants the gift of faith. Effectual grace produces conversion, revival, and awakening. It is particular, given to those whom God has chosen before the foundation of the world (Ephesians 1:4), and it is irresistible in the sense that when God purposes to transform a soul, the transformation occurs.

Grace is the model’s most original element. Four features deserve emphasis:

1. **Exogeneity.** The agent cannot manufacture, earn, or control grace; neither common nor effectual. Even the faith to believe is itself a gift (Ephesians 2:8–9). This is the formal distinction from social capital models where “trust” accumulates endogenously. In our model, the deepest source of transformation comes from outside the system entirely.
2. **Common grace is persistent and universal.** $\bar{G} > 0$ for all agents, always. Even if the agent invests nothing in spiritual growth ($i^S = 0$), the steady-state soul is not zero, it equals $\bar{G}/(\delta^S - \xi)$. This is the “grace floor”: God sustains a minimal spiritual awareness, a residual conscience, even in the most negligent agent. Common grace keeps evil in check and sustains creation. Without it, the low- S equilibrium would collapse into something far worse.
3. **Effectual grace is particular and stochastic.** ε_t^G is not given to all agents in all periods. It represents God’s sovereign decision to regenerate specific hearts at specific times. The timing is not predictable from within the model, but is genuinely exogenous. Christ died so that all who believe in Him would have eternal life (John 3:16), and even the faith to believe is granted by grace, not generated by human will.
4. **Non-negativity.** $\varepsilon_t^G \geq 0$: grace is always a gift, never a punishment. God may allow spiritual decay (through δ^S) but does not actively destroy the soul.

Remark (Common grace and natural law). The observable effects of common grace ($\bar{G} > 0$) closely resemble what the natural law tradition, from Aristotle through Aquinas, attributes to innate moral reason. Both predict that even unregenerate agents exhibit baseline moral behavior: a residual conscience, a sense that theft is wrong, a capacity for elementary cooperation. The economic predictions are equivalent. But the *cause* differs fundamentally. Natural law locates baseline morality in human nature itself, implying an inherent capacity that survives the Fall intact. Common grace locates it in God’s ongoing sustaining act: “He causes his sun to rise on the evil and the good, and sends rain on the righteous and the unrighteous” (Matthew 5:45).

The distinction matters for the model. If baseline morality is inherent, then it is structurally permanent meaning that no force can push S below the natural law floor. If it is sustained by grace, then it depends on God’s continued action, which is freely given but not owed. We adopt the grace interpretation because it preserves divine sovereignty and because it is consistent with the historical observation that moral floors can erode: societies do descend into genocidal evil, which a fixed natural law floor struggles to accommodate.

Remark (Grace and technology shocks). Every economist who reads this will object: “You cannot have an exogenous variable driving your model! That is not a solution: it is an admission of ignorance!” We reply: you already have exogenous variables driving your models. What is TFP? It is a residual, or the part of growth we cannot explain with measured inputs. We call it “technology” and treat it as understood, but it is not. Grace is the spiritual residual. It deserves the same methodological treatment that economists already extend to technology shocks in [Solow \(1956\)](#) and RBC models.

4.6 Optimality conditions

The agent chooses $\{c_t, g_t, i_t^B, i_t^M, i_t^S, \theta_t\}_{t=0}^T$ to maximize (1) subject to (3)–(7). The key first-order conditions are (with full derivations in [Appendix A](#)):

Consumption Euler equation:

$$c_t^{-\sigma(S_t)} = \beta(S_{t+1}) (1 + r_{t+1}) c_{t+1}^{-\sigma(S_{t+1})} \quad (9)$$

Both the discount factor and the curvature depend on S , so the consumption path is jointly determined with the spiritual trajectory.

Giving ratio:

$$\frac{g_t}{c_t} = \psi(S_t)^{1/\sigma(S_t)} \quad (10)$$

When $S = 0$, the agent gives nothing. As S rises, generosity emerges endogenously.

Honesty:

$$\lambda_t \Pi_t = \nu_t \quad (11)$$

where $\nu_t \geq 0$ is the KKT multiplier on the honesty constraint. The left side is the marginal benefit of cheating: one more unit of θ raises income by Π_t , valued at shadow price λ_t . If $\nu_t > 0$, the constraint binds and $\theta_t = \bar{\theta}(1 - S_t)$ where the agent cheats up to the moral limit. If $\nu_t = 0$, the agent chooses not to cheat even when permitted (the future costs of dishonesty, operating through μ_t^S , outweigh the immediate gain).

Equal marginal returns to investment:

$$\mu_t^B \phi^{B'}(i_t^B) = \mu_t^M \frac{\partial \phi^M}{\partial i_t^M} = \mu_t^S \phi^{S'}(i_t^S) = \lambda_t \quad (12)$$

The agent allocates investment across body, mind, and soul until the marginal returns are equalized. This is a standard optimality condition. The novelty is that spiritual investment competes with (and complements) physical and human capital investment.

The shadow value of the soul:

$$\frac{\partial V_{t+1}}{\partial S_{t+1}} = \underbrace{\frac{\partial \beta}{\partial S} \frac{V_{t+2}}{\beta}}_{\text{patience}} + \underbrace{\frac{\partial u}{\partial \sigma} \frac{\partial \sigma}{\partial S}}_{\text{satiation}} + \underbrace{\frac{\partial u}{\partial \psi} \frac{\partial \psi}{\partial S}}_{\text{generosity}} + \underbrace{\nu_{t+1} \bar{\theta}}_{\text{honesty}} \quad (13)$$

This is the central equation of the model. The soul is not just another stock variable; it is a *meta-variable* that transforms preferences, time horizons, social orientation, and moral constraints simultaneously. The four terms in equation (13) define the four channels through which the soul changes everything.

5 Four Channels and Seven Propositions

The soul transforms economic behavior through four channels. Each channel generates testable propositions.

5.1 Channel 1: Patience

The soul transforms the time horizon. A person grounded in Christ thinks long-term; e.g., stewardship, investment, building for the next generation. An agent driven by desire discounts the future heavily and grabs what is available now.

Formally, $\beta(S) = \underline{\beta} + (\bar{\beta} - \underline{\beta})S$ rises with S : when S is high, β is close to 1 and the agent is patient. When S is low, β is low and the agent is impatient. The mechanism has a mainstream precedent: [Becker and Mulligan \(1997\)](#) show that agents can invest resources to raise their discount factor, producing divergent steady states in wealth. Our contribution is to identify the *source* of that investment (devotional practice) and the *shock* that catalyzes it (grace).

Proposition 1 (The Patience Effect). *A higher soul state reduces corruption. Formally, $\partial \theta^* / \partial S^* < 0$ and the expected tenure of corrupt behavior decreases in S .*

Proof sketch. The honesty constraint gives $\theta^* \leq \bar{\theta}(1 - S^*)$, so $\partial \theta^* / \partial S^* = -\bar{\theta} < 0$ when binding. When the constraint is slack, the agent still cheats less at higher S : the discount factor $\beta(S)$ rises with S , increasing the present value of future costs from corruption (detection, punishment, loss of reputation). An impatient agent discounts future punishment heavily and appropriates rents for immediate gain. A patient agent finds the present value of expected penalties exceeds $\lambda_t \Pi_t$ and cheats less even when the moral constraint permits more. Full proof in Appendix B. $\square$

5.2 Channel 2: Satiation

The soul transforms how much we need. A person grounded in Christ is more easily satisfied with material goods; “godliness with contentment is great gain” (1 Timothy 6:6). An agent driven by desire is never satisfied.

Formally, $\sigma(S) = \underline{\sigma} + (\bar{\sigma} - \underline{\sigma})S$ rises with S : higher curvature means the marginal utility of consumption diminishes quickly and the agent saves more. The link between moral identity and consumption restraint is documented by [Bénabou and Tirole \(2011\)](#), who show that agents with

stronger moral self-image resist temptation more effectively, investing in identity as an “asset” that disciplines future behavior.

Proposition 2 (The Satiation Effect). *A higher soul state frees resources from consumption to savings and investment. Formally, $\partial(i^{B*} + i^{M*} + i^{S*})/\partial S^* > 0$.*

Proof sketch. Higher σ reduces marginal utility of consumption for any given c . Higher β tilts the consumption profile toward the future. Together, they imply a flatter consumption path and higher steady-state savings. By the budget constraint, higher savings translate to higher total investment. $\square$

5.3 Channel 3: Generosity

The soul transforms how we treat others. A Christ-centered person gives voluntarily to the poor, to the neighbor, to the stranger. This is not redistribution imposed by the state. It is redistribution that flows naturally from a transformed heart.

Formally, $\psi(S) = \psi_0 S^\eta$ rises with S . The giving-to-consumption ratio $g/c = \psi(S)^{1/\sigma(S)}$ increases monotonically. Andreoni (1990) formalized this mechanism as “warm-glow giving”: agents derive intrinsic utility from the act of giving, independent of the public good produced. Our framework endogenizes Andreoni’s warm glow. It is not a fixed taste parameter but a function of the soul state.

Proposition 3 (Generosity Emergence). *Voluntary giving emerges endogenously from the soul state and is strictly increasing in S . Forced redistribution is unnecessary when S is sufficiently high.*

Proof sketch. From equation (10), $g/c = \psi(S)^{1/\sigma(S)}$ is increasing in S (verified in Appendix B by differentiating $\ln[g/c]$ with respect to S). When S is high enough that voluntary giving exceeds the transfer a social planner would mandate, forced redistribution becomes redundant. $\square$

This proposition has radical policy implications. Government welfare programs, affirmative action schemes, and forced redistribution are all substitutes for generosity in a low-soul society. They are necessary precisely because voluntary giving is insufficient. But they are also inefficient, corruptible, and dignity-reducing. A high-soul society generates the same transfers (and more) through voluntary action, without the bureaucratic overhead or the resentment.

5.4 Channel 4: Honesty

The soul imposes an internal moral constraint. A Christ-centered person *cannot* cheat, not because the law prevents it, but because an internal conviction makes it impossible.

Formally, $\theta_t \leq \bar{\theta}(1 - S_t)$. When $S = 1$, the agent cannot cheat at all. When $S = 0$, the constraint is slack. Mazar et al. (2008) provide direct experimental evidence for this internal ceiling: subjects cheat only up to the point where dishonesty would force a negative revision of their moral self-concept. Our soul state S is the variable that determines where that ceiling sits.

Proposition 4 (The Honesty Dividend). *Aggregate dishonesty reduces total factor productivity. When agents have higher S , aggregate TFP rises.*

Proof sketch. While corruption benefits the individual (equation 3), it diverts resources from productive use at the aggregate level. A corrupt official who appropriates infrastructure funds raises her own consumption but destroys the public good. Aggregate effective labor is therefore $\bar{N} = \int M_i B_i (1 - \theta_i) di$, where θ_i captures the productivity loss from each agent's dishonesty. Aggregate output is $Y = A \bar{N}^\alpha K^{1-\alpha}$. Measured TFP (the Solow residual, computed from observed inputs $N = \int M_i B_i di$) equals $A \cdot (1 - \bar{\theta}(1 - \bar{S}))^\alpha$. The honesty dividend is:

$$\Delta \text{TFP} \approx \alpha \bar{\theta} \Delta \bar{S} \quad (14)$$

With $\alpha = 0.65$ and $\bar{\theta} = 0.3$, a one-unit increase in average soul state raises measured TFP growth by approximately 20%. $\square$

5.5 Additional propositions

Proposition 5 (The Forgiveness Equilibrium). *In a two-group setting, higher S in both groups makes mutual forgiveness the unique Nash equilibrium. Below a threshold S^F , revenge is the dominant strategy.*

Proof sketch. In a two-player game with strategies Forgive (F) and Revenge (R), the payoff from mutual forgiveness $\pi^{FF}(S) = \beta(S)V^{\text{peace}} + \psi(S)\Omega$ is increasing in S (patience values peace; generosity values the other's welfare). The payoff from unilateral revenge $\pi^{RF}(S) = R_0 - \psi(S)\Gamma$ is decreasing in S (generosity penalizes harming others). By the intermediate value theorem, there exists a unique threshold S^F where $\pi^{FF} = \pi^{RF}$. Above S^F , forgiveness is dominant. Below it, revenge is dominant. $\square$

Proposition 6 (The Unemployment–Soul Nexus). *In a Nash bargaining framework, soul-dependent markups reduce structural unemployment. Formally, $\partial \bar{u} / \partial \bar{S} < 0$.*

Proof sketch. Structural unemployment is $\bar{u} = 1 - (1 - \tau) / [(1 + \mu^P)(1 + \mu^W)]$. Unions with higher S internalize the cost of unemployment on others (generosity channel) and moderate wage demands: $\mu^W(S) = \frac{\beta_{\text{union}}}{1 - \beta_{\text{union}}} (1 - \psi(S)\kappa)$. Firms with higher S reduce price markups: $\mu^P(S) = \mu_0^P (1 - S\zeta)$. Since $\bar{u}$ is increasing in both markups and both are decreasing in S , the result follows. $\square$

Proposition 7 (Critical Mass and Multiple Equilibria). *There exist two stable steady states: a low-soul equilibrium (high corruption, low trust, low growth) and a high-soul equilibrium (low corruption, high trust, high growth). The transition requires a critical mass of agents with $S > S^*$.*

Proof sketch. With social interaction (peer effects $\xi \bar{S}_t$ in the soul law of motion), the steady-state condition becomes $(\delta^S - \xi) \bar{S}^* = \phi^S(i^{S^*}(\bar{S}^*)) + \bar{G}$. The right side is convex in $\bar{S}$ (investment complementarity: it is easier to invest in soul when surrounded by others doing the same). This convexity generates multiple crossings with the linear left side; two stable equilibria separated by an unstable tipping point. $\square$

6 Two Cities, Two Equilibria

Augustine’s *De Civitate Dei* was written between 413 and 426 AD, as Rome was sacked by the Visigoths and pagans blamed Christianity for the Empire’s decline. Augustine’s response was not to defend Rome but to reframe the question. Rome, he argued, was a city built on the love of self, or the *libido dominandi*, the lust for domination. Its decline was not caused by Christianity but by the same disordered desires that built it. The City of God, by contrast, is built on the love of God and of neighbor for God’s sake. It exists alongside the City of Man in every generation, often invisible, but it is the only city that endures (Augustine, 426).

The parallels with our model are exact:

	City of Man ($\bar{S}_L$)	City of God ($\bar{S}_H$)
Driving force	Self-love (<i>amor sui</i>)	Love of God (<i>amor Dei</i>)
Desire	Unbounded, insatiable	Bounded by contentment
Time horizon	Short-term, impatient	Long-term, patient
Social orientation	Self-interest, exploitation	Generosity, service
Moral constraint	External only (law, punishment)	Internal (conscience, conviction)
Model parameter	Low S , low β , low σ , $\psi \approx 0$	High S , high β , high σ , high ψ
Corruption	High ($\theta \approx \bar{\theta}$)	Low ($\theta \approx 0$)
Savings	Low	High
Forgiveness	Revenge dominant	Forgiveness dominant
Growth	Stagnant	Flourishing

Augustine understood what our model formalizes: that the difference between a flourishing society and a decaying one is not primarily institutional but *dispositional*. It depends on what the citizens love. “Two cities have been formed by two loves: the earthly by the love of self, even to the contempt of God; the heavenly by the love of God, even to the contempt of self” (*De Civitate Dei*, XIV.28).

Kierkegaard sharpens the diagnosis. In *The Sickness Unto Death*, he defines despair as the condition of a self that “does not will to be itself” or, equivalently, that “wills not to be the self that it truly is” (Kierkegaard, 1849). For Kierkegaard, the self is constituted by a Power outside itself: God. Despair is the rejection of that relationship: the self tries to construct itself on its own terms, becoming a fabrication, a mere image that is not the person God intended. In the language of our model, this is precisely the low- S agent. When $S \rightarrow 0$, the agent rejects the divine constitution of the self and builds an identity around desire (power, status, wealth). The result is not freedom but bondage: insatiable consumption (σ low), impatience (β low), dishonesty (θ high), and indifference to others ($\psi \approx 0$). The agent becomes less than human, not more. And when an entire society is composed of such agents, it is easy to see why it turns to evil, that ultimately destroys itself. The low- S equilibrium is not merely an economic trap. It is, in Kierkegaard’s sense, a society in despair: a collective of selves that have rejected the Power that constituted them, and in doing so have lost the capacity to flourish.

6.1 The critical mass and the role of the Church

Proposition 7 shows that the transition between equilibria requires a critical mass. In our calibration, this threshold is approximately $\bar{S}^* \approx 0.65$, roughly two-thirds of the population must have a soul state above the tipping point for the social multiplier to pull the entire society upward.

This has implications for the Church. The Church is the institution through which grace is proclaimed, fellowship sustained, and devotional investment (i^S) made most productive. The social multiplier ξ is highest within church communities where weekly worship, small groups, accountability structures, shared Scripture all amplify the individual's spiritual growth. The model predicts that a society's trajectory depends critically on the size, vitality, and theological seriousness of its churches.

6.2 Historical evidence: Rwanda and South Africa

Rwanda crossed the threshold. After the genocide of 1994, the Rwandan people faced a choice: perpetual revenge or radical forgiveness. The Gacaca courts (12,000 community tribunals processing 1.2 million cases) were the mechanism. But the mechanism worked because there was a spiritual transformation underneath it. Rwanda experienced a Christian revival in the years following the genocide. Churches became centers of reconciliation. Perpetrators confessed not just to human judges but to God. Victims forgave not because it was politically expedient but because Christ commanded it. Rwanda's economy has grown at 7–8% per year since 2000.

South Africa had the Truth and Reconciliation Commission. It was a noble attempt. But the TRC operated at the level of mind (rational inquiry, testimony, amnesty) without reaching the soul. South Africa heard the truth but did not forgive. Anger did not dissipate. The result was redistribution-as-revenge: policies designed not to heal but to transfer, breeding corruption, resentment, and economic stagnation. In the language of our model, South Africa's average soul state $\bar{S}$ remains below the critical threshold S^F .

7 Numerical Simulations

This section presents calibrated simulations illustrating the model's quantitative predictions. The baseline calibration is summarized in Table 2.

7.1 Comparative statics of the soul state

Figure 1 shows how steady-state outcomes change as S varies from near zero to near one.

Panel (c) is perhaps the most striking. At $S = 0$, the agent consumes virtually everything. At $S = 0.8$, investment absorbs nearly half of income. This is the micro-foundation for the claim that spiritual transformation drives capital accumulation. The soul does not compete with growth, but rather *enables* it.

Table 2: Baseline calibration parameters

Parameter	Interpretation	Value	Source/Justification
β	Min patience (low soul)	0.92	Standard macro
$\bar{\beta}$	Max patience (high soul)	0.98	Standard macro
$\underline{\sigma}$	Min curvature	1.2	Standard CRRA
$\bar{\sigma}$	Max curvature	3.0	Higher = more easily satisfied
ψ_0	Scale of giving weight	1.0	Matches tithing data
η	Convexity of giving in S	1.0	Linear response
$\bar{\theta}$	Max feasible cheating	0.30	Corruption surveys
δ^S	Spiritual decay rate	0.30	Moderate decay from neglect
$\bar{G}$	Persistent grace	0.05	Small but positive
ξ	Social multiplier	0.10	Peer effects literature
α	Labor share	0.65	National accounts

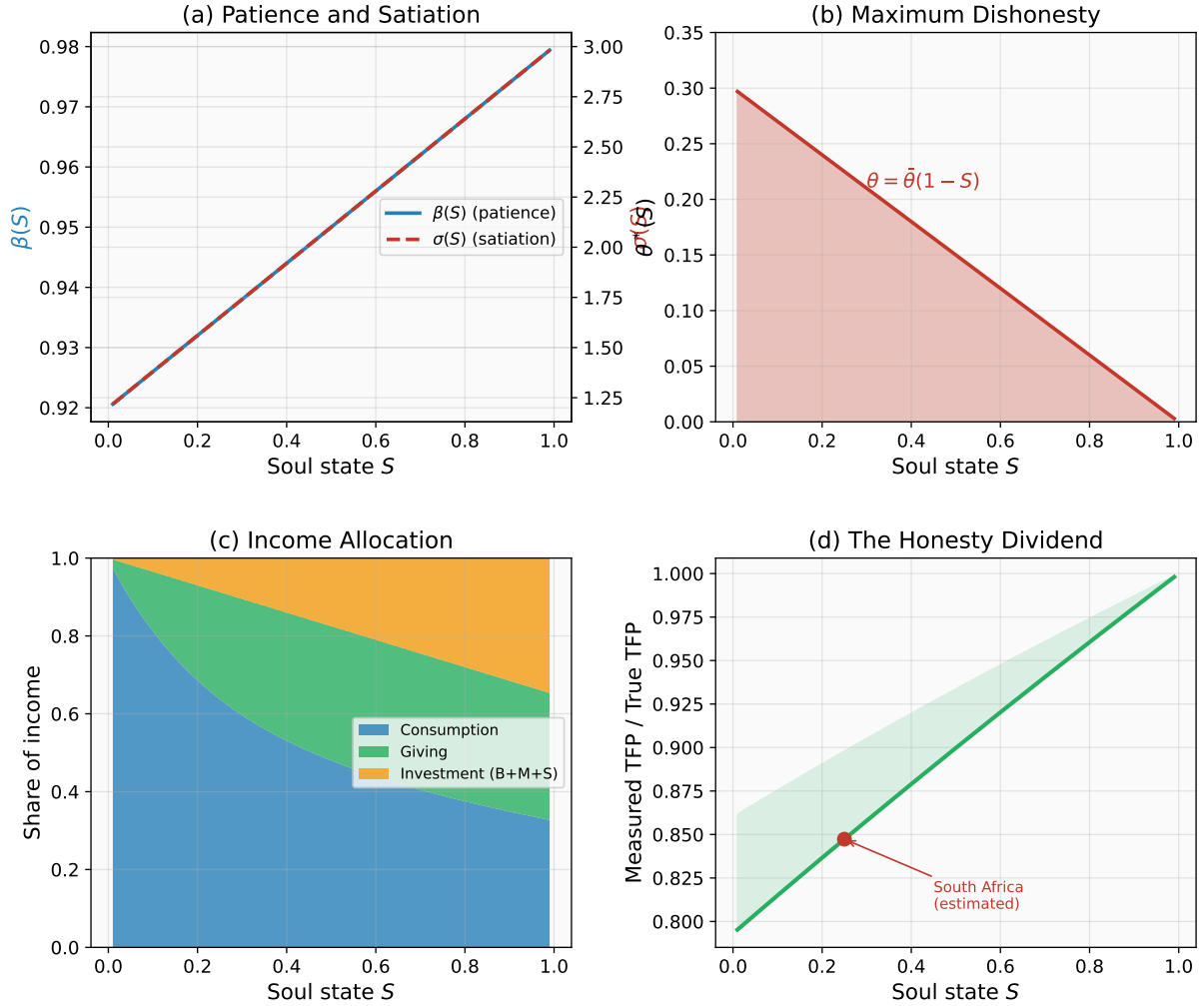
Figure 1: Comparative Statics of the Soul State

Figure 1: Comparative statics of the soul state. Panel (a): patience $\beta(S)$ and satiation $\sigma(S)$ rise with S . Panel (b): maximum dishonesty $\theta^*(S) = \bar{\theta}(1 - S)$ falls. Panel (c): the allocation of income shifts from consumption toward giving and investment as S rises. Panel (d): measured TFP rises as honesty improves (the honesty dividend).

7.2 The two-equilibria phase diagram

Figure 2 plots the steady-state condition from Proposition 7. The blue line represents net depreciation ($\delta^S \bar{S}$, linear). The red curve represents total replenishment (grace + investment response + social multiplier). The replenishment curve is S-shaped: flat at low $\bar{S}$ (nobody invests in soul when the social environment is hostile), then steep (complementarity kicks in), then flattening (diminishing returns).

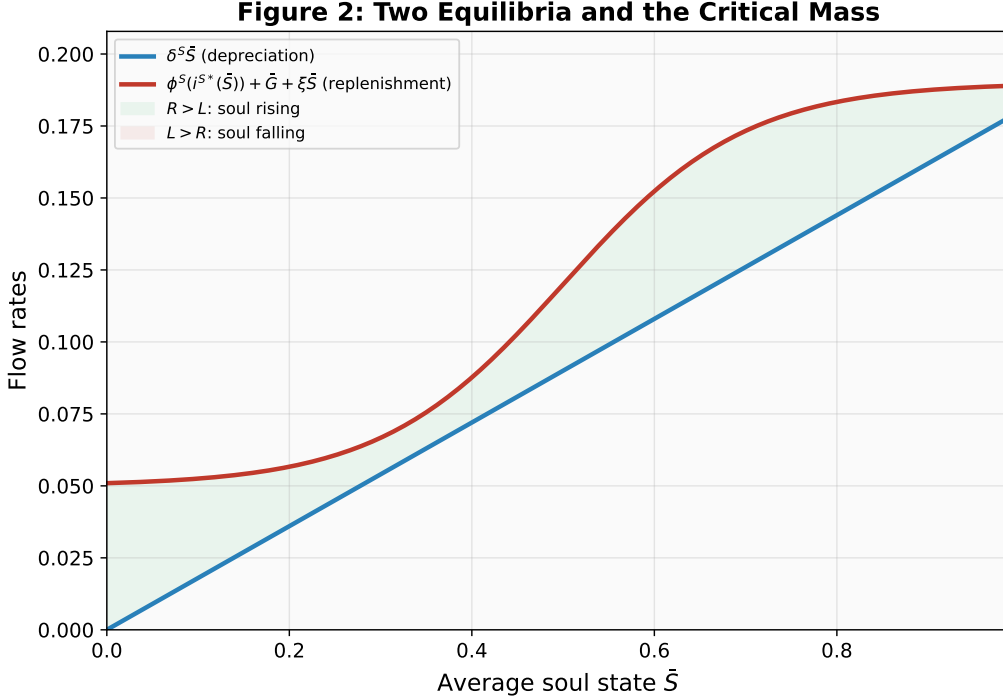


Figure 2: Phase diagram for the average soul state $\bar{S}$. Two stable equilibria emerge: a low-soul trap ($\bar{S}_L \approx 0.28$) and a high-soul flourishing state ($\bar{S}_H \rightarrow 1$). The critical mass $\bar{S}^* \approx 0.65$ separates the basins of attraction.

7.3 The welfare landscape

Figure 3 maps steady-state welfare as a function of grace ($\bar{G}$) and devotional investment (i^S).

The key insight: grace without devotion (moving right along the bottom edge) raises welfare only modestly. Devotion without grace (moving up along the left edge) is even less effective. The interaction is the key: grace opens the heart, devotion builds the habit, and together they transform the agent. This is the economic content of “work out your salvation with fear and trembling, for it is God who works in you” (Philippians 2:12–13).

7.4 GDP decomposition by soul channel

Figure 4 decomposes the GDP gain from higher $\bar{S}$ into three channels: the honesty dividend (green), human capital from patience (blue), and physical capital from savings (gold).

The composition shifts with S . At low S , the honesty dividend dominates; eliminating corruption is the single biggest gain. At moderate S , human capital takes over as patient agents

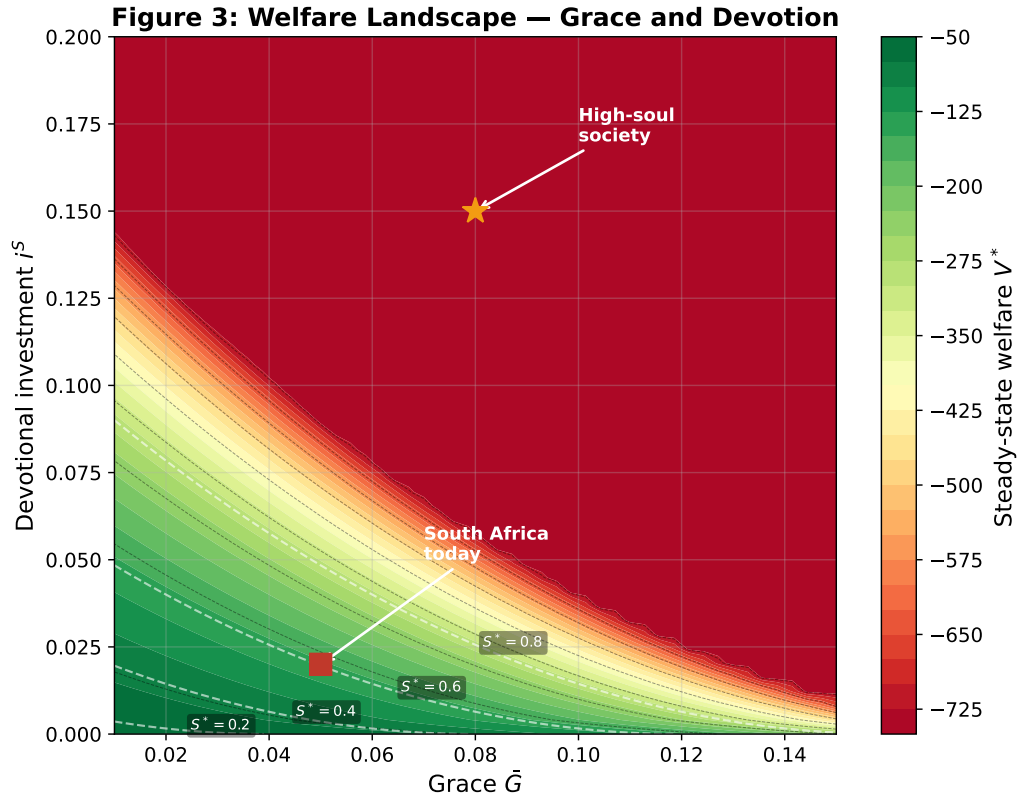


Figure 3: Welfare landscape over grace and devotional investment. Grace and devotion are complements, not substitutes. The welfare gap between a low-soul and high-soul society spans orders of magnitude.

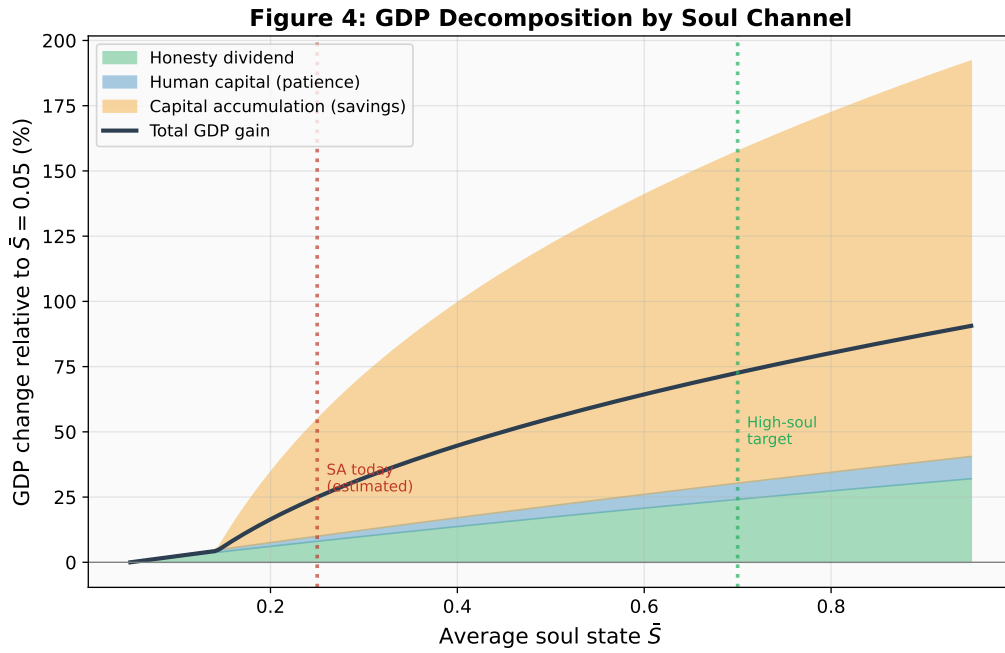


Figure 4: GDP decomposition by soul channel, relative to a baseline of $\tilde{S} = 0.05$. Moving from $\tilde{S} \approx 0.25$ to $\tilde{S} = 0.70$ would raise GDP by approximately 50–70%.

invest in education. At high S , capital accumulation compounds where high savings rates build the capital stock.

7.5 The forgiveness game

Figure 5 plots the payoff functions from Proposition 5.

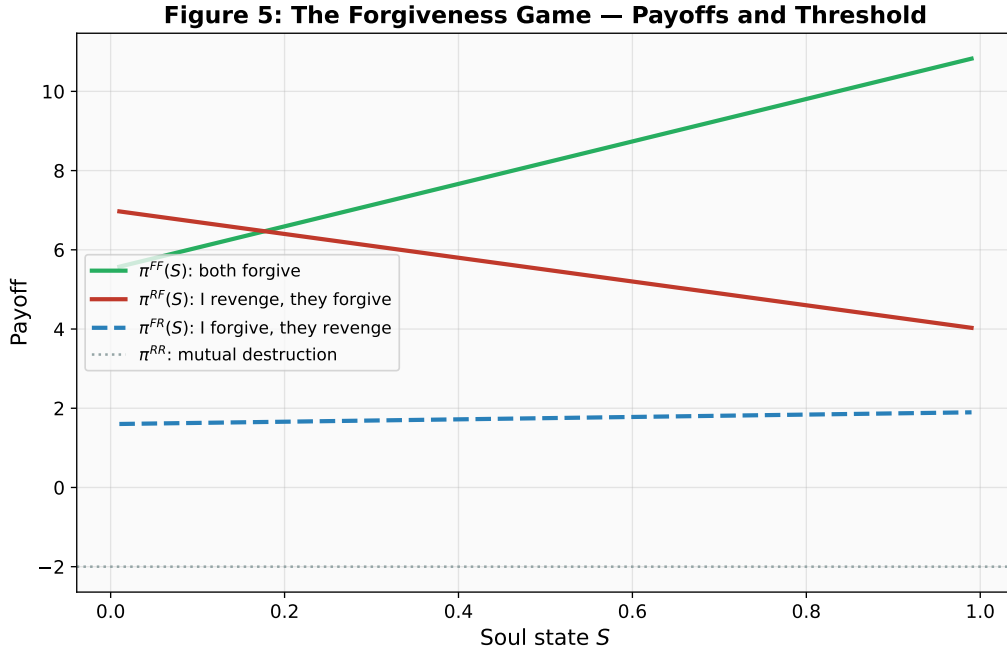


Figure 5: Payoffs in the forgiveness game. At the threshold $S^F \approx 0.25$, mutual forgiveness becomes the dominant strategy. Below it, revenge dominates.

The calibration reveals a sobering pattern. At $S = 0$, revenge is overwhelmingly attractive. This is the world of vendetta and “our turn to eat.” But above S^F , the gap widens rapidly: at $S = 0.8$, revenge is not just inferior, it is unthinkable for a high- S agent. Forgiveness is not weakness. It is the optimal strategy of the transformed soul.

7.6 Transition dynamics

Figure 6 simulates a society starting in the low equilibrium ($\bar{S}_0 = 0.20$) that experiences a grace-driven revival beginning in period 20.

The transition exhibits the classic S-curve: slow at first (the low- S trap resists change), then accelerating (investment complementarity kicks in), then plateauing. GDP rises by approximately 30% over 60 periods. This is consistent with post-revival economies: South Korea in the 1960s–90s (where Christianity grew from 2% to 30% of the population alongside the economic miracle), Rwanda in the 2000s–present, and the Wesleyan revival in 18th-century England that preceded the Industrial Revolution.

7.7 Summary of simulation results

The magnitude should give pause. No tax incentive, no regulatory framework, no constitutional amendment can produce effects of this magnitude. The model says what Augustine said: the

Figure 6: Transition Dynamics — A Grace-Driven Revival

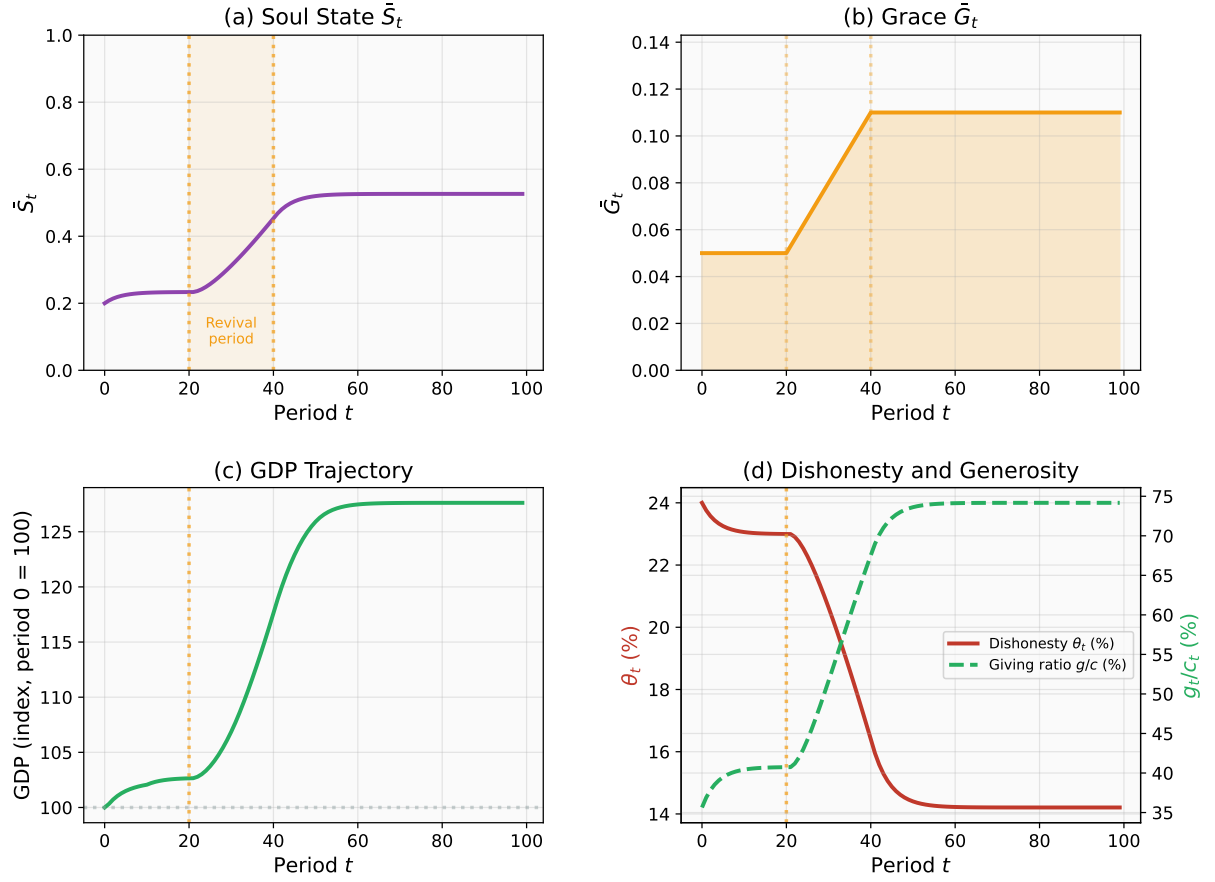


Figure 6: Transition dynamics following a grace-driven revival. The soul state follows an S-curve. GDP rises by approximately 30% over 60 periods. Dishonesty falls while the giving ratio rises. The transition takes a generation.

Table 3: Model outcomes at low and high soul states

Outcome	Low S (≈ 0.20)	High S (≈ 0.70)
Dishonesty rate θ	24%	9%
Giving/consumption ratio	$\sim 5\%$	$\sim 65\%$
Investment share of income	$\sim 15\%$	$\sim 45\%$
Measured TFP (fraction of true)	85%	96%
GDP (index, low $S = 100$)	100	~ 160
Dominant strategy	Revenge	Forgiveness

binding constraint on a society is not its institutions but the disposition of its citizens' hearts.

8 Nesting Result: Homo Economicus as a Special Case

The trichotomous model does not reject mainstream economics. It contains it.

Claim: When $S_t = 0$ for all t , the trichotomous agent reduces to the standard utility-maximizing agent of neoclassical economics.

Proof. Setting $S_t = 0$:

- $\beta(0) = \underline{\beta}$: a constant discount factor.
- $\sigma(0) = \underline{\sigma}$: a constant curvature (standard CRRA utility).
- $\psi(0) = 0$: no weight on giving. The agent is purely self-interested.
- Honesty constraint: $\theta_t \leq \bar{\theta}$. Maximally slack.
- Soul law of motion: $S_{t+1} = \mathcal{G}_t$. The soul is sustained only by grace.

The optimization problem reduces to:

$$\max \mathbb{E}_0 \sum_{t=0}^T \underline{\beta}^t \frac{c_t^{1-\underline{\sigma}}}{1-\underline{\sigma}} \quad \text{s.t.} \quad c_t + i_t^B + i_t^M \leq w_t M_t B_t + \bar{\theta} \Pi_t \quad (15)$$

This is a textbook lifecycle model with human capital accumulation (Ben-Porath, 1967), health investment, and rent-seeking. The agent cheats to the maximum extent ($\theta = \bar{\theta}$) and has no reason not to, i.e., there is no internal moral constraint and no weight on the welfare of others. The soul investment term disappears because $\mu^S = 0$ when S has no value. $\square$

This nesting result is important for two reasons. First, it means that any empirical test of the trichotomous model against the standard model is a test of $S > 0$ versus $S = 0$; it is a test of whether the soul matters, not whether the standard model is wrong. Second, it means that the trichotomous model is *at least as good* as the standard model in every application where the soul is irrelevant. We lose nothing by extending the framework and gain the ability to explain phenomena such as persistent corruption, the failure of institutional reform, the power of forgiveness, that the standard model cannot.

9 A Challenge to the Secular Economist

We address the skeptical reader directly. Three objections are likely.

9.1 “Norms can explain this without God”

You may prefer to call it norms rather than soul. Social norms, cultural capital, civic virtue are familiar terms and comfortable. And you are not wrong that norms matter. Knack and Keefer (1997) show that trust predicts growth. Putnam (1993) demonstrates that civic engagement underpins institutional quality.

But you must answer: *where do norms come from?*

If you say evolution, i.e., that cooperation emerged because cooperative groups outcompeted selfish ones, you have explained why cooperation can *persist*, but not why it should be *valued*. Evolution explains survival, not morality. An evolutionary account of cooperation is compatible with a society that cooperates instrumentally while remaining indifferent to justice, mercy, and truth. Ants cooperate beautifully. They do not forgive.

If you say culture, i.e., that norms are transmitted through families, schools, and institutions, you have pushed the question back one step. Where did the culture come from? Why do some cultures produce trust and others produce corruption? [Acemoglu and Robinson \(2012\)](#) argue that institutions are the answer, and [North \(1990\)](#) provides the foundational framework. But institutions are designed and operated by people, and the same institutional blueprint produces radically different outcomes in different societies. Culture is not an explanation. It is a description of the thing that needs explaining.

If you say reason, i.e., that rational agents can derive moral norms from first principles, you face Hume’s guillotine: you cannot derive an “ought” from an “is.” Reason can tell you that cooperation is efficient. It cannot tell you that you *should* cooperate when defection is more profitable.

9.2 “The model is not falsifiable”

Grace is exogenous and stochastic. How can you test a model whose key variable is unobservable and uncontrollable?

We reply: the model generates *observable* predictions. It predicts that societies with higher church attendance, controlling for income, will have lower corruption (Proposition 1), higher savings (Proposition 2), more voluntary charity (Proposition 3), higher TFP (Proposition 4), and lower structural unemployment (Proposition 6). It predicts that post-revival societies will exhibit S-curve transitions in economic indicators. It predicts that institutional reform without spiritual transformation will fail.

These are testable predictions. The empirical literature already provides suggestive evidence: [Barro and McCleary \(2003\)](#) find that religious beliefs (particularly belief in hell) predict economic growth; [Guiso et al. \(2006\)](#) show that Christian denomination affects economic attitudes; and the contrast between Rwanda (spiritual revival followed by economic transformation) and South Africa (institutional reform without spiritual transformation, followed by stagnation) is a natural experiment in the model’s key mechanism.

9.3 “You have a reverse causality problem”

Does higher S cause prosperity, or does prosperity cause higher S ? This is a legitimate concern. Three responses.

First, the *timing* evidence favors our direction. [Woodberry \(2012\)](#) demonstrates that conversionary Protestant missions predict democracy, literacy, and economic development decades later, and the missions preceded the development, not the reverse. South Korea’s Christian population grew from 2% in 1945 to 30% by 1990, and the growth in Christianity *preceded* the economic take-off, not the other way around. The Wesleyan revival in 18th-century England preceded the Industrial Revolution by decades. Rwanda’s Christian revival preceded, not followed,

its 7–8% GDP growth.

Second, the model generates *distinguishing predictions* that the reverse-causality story cannot match. If prosperity causes religiosity, we should see rich countries becoming more religious over time. The opposite is happening in Western Europe. If prosperity causes forgiveness, we should see rich societies forgiving historical wrongs more easily. We do not—wealthy, secular Western Europe has not forgiven more readily than poor, Christian Rwanda. The model’s specific predictions about forgiveness equilibria, voluntary giving replacing state redistribution, and S-curve transitions following grace shocks have no natural reverse-causality explanation.

Third, the nesting result provides a *falsification criterion*. If $S = 0$ and the standard model is sufficient, then institutional quality alone should predict economic outcomes after controlling for capital, labor, and technology. The persistent failure of well-designed institutions in low-trust environments such as South Africa’s Constitution, the Chapter 9 institutions, an independent central bank is evidence that the standard model is incomplete. Something is missing. We identify that something as the soul.

9.4 “You are smuggling theology into economics”

Yes. And you are smuggling atheism into economics by assuming $S = 0$. The difference is that we are explicit about our assumption and you are not. Every model makes metaphysical commitments. The standard model assumes that the agent is fully described by preferences over consumption and leisure, that these preferences are fixed, and that moral considerations enter only through external constraints. These are not value-neutral assumptions. They are the assumptions of a particular worldview; one that excludes the soul by fiat, not by evidence.

We propose a more general framework. If you believe $S = 0$, our model reduces to yours and you lose nothing. If $S > 0$, the model explains phenomena that yours cannot. The question is not whether we are smuggling theology into economics. The question is whether the data is better explained with or without the soul.

9.5 “This is a god-of-the-gaps argument”

The objection runs as follows: you cannot explain why some societies flourish and others decay, so you insert God into the residual and call it grace. This is the “god of the gaps”—an argument from ignorance dressed in equations.

We take this objection seriously. The god-of-the-gaps fallacy is real: attributing the currently unexplained to divine action, only to retreat when science advances, is bad theology and worse science. But that is not what we do. Our argument is not “we cannot explain the residual, therefore God.” Our argument is: here is a more general model that nests the standard one, generates additional testable predictions, and fits the data better on specific margins. The nesting result (Section 8) is the key: if $S = 0$ is sufficient, reject our model. We have given you the falsification criterion. A god-of-the-gaps argument provides no such criterion.

Consider the parallel. When Solow (1956) attributed the majority of growth to an unexplained residual and called it “technology,” no one accused him of a “technology of the gaps” argument. When Real Business Cycle theorists modeled productivity as an exogenous stochastic process, no one demanded that they first explain where technology shocks come from before

being allowed to study their consequences. We model grace as an exogenous stochastic process and study its consequences. The methodology is identical. The discomfort is not scientific. It is metaphysical.

But we go further than methodological parity. We make a specific claim about the *source* and *mechanism* of grace that no secular residual can match. The claim is this: grace enters the world through the cross of Christ.

The cross is the model’s deepest exogenous intervention. In Christ, God enters the system from outside, not as a metaphor, not as a cultural force, but as a historical person who lived, died under Roman execution, and rose from the dead. The crucifixion is the event in which the just penalty for human rejection of God (the very rejection that produces the low- S equilibrium) is borne by God himself. The resurrection is the demonstration that the penalty has been fully paid. And the gift that flows from this event—grace, unearned and unearnable—is what makes spiritual transformation possible. Christ died so that all who believe in Him would have eternal life. And even the faith to believe is not the agent’s own doing: “For by grace you have been saved through faith. And this is not your own doing; it is the gift of God, not a result of works, so that no one may boast” (Ephesians 2:8–9). Grace is sovereign from first to last. The decision to bestow it was made before the foundation of the world (Ephesians 1:4). Grace is exogenous not merely contemporaneously but *eternally*.

The cross answers both objections simultaneously:

1. **It is not a gap-filler.** The cross is a specific historical claim with specific, testable consequences. If Christ did not rise, the entire mechanism collapses (1 Corinthians 15:14). A god-of-the-gaps argument retreats from evidence. The Christian claim *invites* it.
2. **Grace is universal *and* particular.** “For God so loved the *world*” (John 3:16); this love is expressed in common grace, which sustains all of creation. In the model, $\bar{G} > 0$ for all agents, always. No society is abandoned. The rain falls on the just and the unjust. But *effectual* grace, the transformative shock ε_t^G that regenerates the heart, is particular. Christ died so that all who believe in Him would have eternal life, and even the faith to believe is itself a gift of grace (Ephesians 2:8–9), granted to those whom God chose before the foundation of the world (Ephesians 1:4). The agent cannot generate ε_t^G from within. This is not a restriction on God’s generosity. It is the deepest expression of His sovereignty: that salvation, from first to last, is His work and not ours.

This leads to a final point about the low- S trap. Common grace ($\bar{G}$) keeps even the lowest- S society from total collapse; it sustains a residual conscience, a minimal moral order, a dim awareness that corruption is wrong even where it is practiced. But common grace alone cannot lift a society across the threshold S^* . That requires effectual grace: the stochastic shocks ε_t^G that transform individual hearts, producing agents whose patience, honesty, generosity, and satiation generate the social multiplier $\xi \bar{S}_t$ that pulls others upward. The low- S equilibrium persists not because God withholds common grace—He does not—but because effectual grace has not yet reached the critical mass. “Light has come into the world, and people loved the darkness rather than the light” (John 3:19). And the transition to the high equilibrium requires what the model predicts: a critical mass of agents in whom effectual grace has worked, whose

transformed behavior generates the social multiplier that pulls the society across the threshold. This is, in economic language, what the New Testament calls the Church.

10 Conclusion

We have presented a formal economic model in which the agent is composed of body, mind, and soul. The soul transforms the utility function through four channels: patience, satiation, generosity, and honesty. Grace, an exogenous gift from God, provides the mechanism for spiritual transformation that human effort alone cannot produce. The model nests standard homo economicus as the special case $S = 0$ and generates seven testable propositions connecting the soul state to observable economic outcomes.

The model's deepest claim is also its most controversial: that the binding constraint on human flourishing is not capital, not technology, not institutions, but the disposition of the human soul. Augustine said it in the fifth century. We have said it in the language of dynamic optimization.

This is not a paper that will satisfy everyone. The secular economist will object to the exogeneity of grace. The theologian may object to the formalization of the sacred. The policy maker may wonder what to *do* with a model whose key variable cannot be legislated.

To the economist, we say: you already model exogenous variables you cannot fully explain. Grace deserves the same treatment.

To the theologian, we say: formalization is not reduction. The model does not claim to capture the fullness of God's grace. It claims only to capture its *economic shadow*, or the way it transforms behavior, incentives, and outcomes in the measurable world.

To the policy maker, we say: you cannot legislate grace, but you can stop obstructing it. Protect religious freedom. Support churches and faith communities. Design institutions that reward virtue rather than gaming. And understand that the deepest reforms, the ones that actually transform a society, come not from Pretoria or Washington or Brussels, but from the human heart, touched by a power that no economist can endogenize.

Augustine ended *De Civitate Dei* with a vision of the eternal Sabbath: the rest that comes when the pilgrim city finally reaches its home. We end with the more modest hope that the tools of economics, honestly applied, can point in the same direction.

Soli Deo Gloria.

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A Derivations

A.1 The Lagrangian

The agent maximizes:

$$V_0 = \mathbb{E}_0 \sum_{t=0}^T \beta(S_t)^t u(c_t, g_t; S_t)$$

subject to:

$$\begin{aligned} c_t + g_t + i_t^B + i_t^M + i_t^S &= w_t M_t B_t + \theta_t \Pi_t && \text{(Budget)} \\ \theta_t &\leq \bar{\theta}(1 - S_t) && \text{(Honesty)} \\ B_{t+1} &= (1 - \delta_t^B) B_t + \phi^B(i_t^B) && \text{(Body)} \\ M_{t+1} &= (1 - \delta_t^M) M_t + \phi^M(i_t^M, M_t) && \text{(Mind)} \\ S_{t+1} &= (1 - \delta_t^S) S_t + \phi^S(i_t^S) + \mathcal{G}_t && \text{(Soul)} \end{aligned}$$

The Lagrangian is:

$$\begin{aligned} \mathcal{L} = \mathbb{E}_0 \sum_{t=0}^T & \left\{ \beta(S_t)^t \left[\frac{c_t^{1-\sigma(S_t)}}{1-\sigma(S_t)} + \psi(S_t) \frac{g_t^{1-\sigma(S_t)}}{1-\sigma(S_t)} \right] \right. \\ & - \lambda_t [c_t + g_t + i_t^B + i_t^M + i_t^S - w_t M_t B_t - \theta_t \Pi_t] \\ & - \nu_t [\theta_t - \bar{\theta}(1 - S_t)] \\ & - \mu_t^B [B_{t+1} - (1 - \delta_t^B) B_t - \phi^B(i_t^B)] \\ & - \mu_t^M [M_{t+1} - (1 - \delta_t^M) M_t - \phi^M(i_t^M, M_t)] \\ & \left. - \mu_t^S [S_{t+1} - (1 - \delta_t^S) S_t - \phi^S(i_t^S) - \mathcal{G}_t] \right\} \end{aligned} \quad (16)$$

A.2 First-order conditions

Consumption:

$$\frac{\partial \mathcal{L}}{\partial c_t} = \beta(S_t)^t \cdot c_t^{-\sigma(S_t)} - \lambda_t = 0 \implies c_t^{-\sigma(S_t)} = \tilde{\lambda}_t$$

where $\tilde{\lambda}_t \equiv \lambda_t / \beta(S_t)^t$ is the current-value multiplier.

Giving:

$$\psi(S_t) \cdot g_t^{-\sigma(S_t)} = \tilde{\lambda}_t$$

Setting equal with the consumption FOC and rearranging:

$$\left(\frac{g_t}{c_t} \right)^\sigma = \psi(S_t) \implies \frac{g_t}{c_t} = \psi(S_t)^{1/\sigma(S_t)}$$

Investment in body, mind, and soul:

$$\lambda_t = \mu_t^B \phi^{B'}(i_t^B) \quad (17)$$

$$\lambda_t = \mu_t^M \frac{\partial \phi^M}{\partial i_t^M} \quad (18)$$

$$\lambda_t = \mu_t^S \phi^{S'}(i_t^S) \quad (19)$$

At the optimum, the marginal cost of investment (λ_t) equals the marginal product times the shadow value for each type of capital.

Honesty:

$$\lambda_t \Pi_t = \nu_t$$

The marginal benefit of cheating ($\lambda_t \Pi_t$: one more unit of θ raises income by Π_t , valued at shadow price λ_t) equals the shadow cost of the moral constraint ($\nu_t \geq 0$). When $\nu_t > 0$, the constraint binds and $\theta_t = \bar{\theta}(1 - S_t)$.

A.3 Co-state equations

Shadow value of body:

$$\mu_t^B = \beta(S_{t+1}) [\lambda_{t+1} w_{t+1} M_{t+1} + \mu_{t+1}^B (1 - \delta_{t+1}^B)] \quad (20)$$

Shadow value of mind:

$$\mu_t^M = \beta(S_{t+1}) \left[\lambda_{t+1} w_{t+1} B_{t+1} + \mu_{t+1}^M \left((1 - \delta^M) + \frac{\partial \phi^M}{\partial M_{t+1}} \right) \right] \quad (21)$$

Note that with the corrected budget constraint, the shadow values of body and mind depend on legitimate earnings $w_t M_t B_t$ (not reduced by θ). The honesty choice affects income through the $\theta_t \Pi_t$ rent term, not through the productivity of body and mind. The term $\partial \phi^M / \partial M$ is the self-productivity of human capital (Ben-Porath effect): existing knowledge helps acquire new knowledge.

Shadow value of soul:

$$\mu_t^S = \beta(S_{t+1}) \left[\frac{\partial V_{t+1}}{\partial S_{t+1}} + \mu_{t+1}^S (1 - \delta^S) \right] \quad (22)$$

where the total marginal value of the soul collects all four channels:

$$\frac{\partial V_{t+1}}{\partial S_{t+1}} = \underbrace{\frac{\partial \beta}{\partial S} \frac{V_{t+2}}{\beta}}_{\text{patience}} + \underbrace{\frac{\partial u}{\partial \sigma} \frac{\partial \sigma}{\partial S}}_{\text{satiation}} + \underbrace{\frac{\partial u}{\partial \psi} \frac{\partial \psi}{\partial S}}_{\text{generosity}} + \underbrace{\nu_{t+1} \bar{\theta}}_{\text{honesty}} \quad (23)$$

The honesty term $\nu_{t+1} \bar{\theta}$ is positive when the constraint binds. Higher S tightens the constraint, but the soul's value accounts for this through the other channels. The $\nu \bar{\theta}$ term captures the economic value of integrity—a commitment device.

A.4 Steady-state conditions

In steady state ($B_{t+1} = B_t$, $M_{t+1} = M_t$, $S_{t+1} = S_t$):

$$\delta^B B^* = \phi^B(i^{B*}) \quad (24)$$

$$\delta^M M^* = \phi^M(i^{M*}, M^*) \quad (25)$$

$$\delta^S S^* = \phi^S(i^{S*}) + \bar{G} \quad (26)$$

The grace floor: if $i^{S*} = 0$, then $S_{\min}^* = \bar{G}/\delta^S$ for the individual agent. In equilibrium, when the social multiplier is included ($S^* = \bar{S}^*$), the grace floor becomes $\bar{G}/(\delta^S - \xi)$, which is higher: the community sustains a baseline soul state above what any isolated individual could maintain. Even a completely negligent agent maintains a non-zero soul state because God's grace persists.

A.5 The consumption Euler equation

From the FOCs at t and $t + 1$:

$$c_t^{-\sigma(S_t)} = \beta(S_{t+1})(1 + r_{t+1}) c_{t+1}^{-\sigma(S_{t+1})} \quad (27)$$

Two departures from the standard Euler equation: (1) the discount factor $\beta(S_{t+1})$ depends on the soul—a more spiritual agent is more patient; (2) the curvature $\sigma(S_t)$ differs across periods if S changes—the agent becomes progressively more easily satisfied.

Log-linearizing with S approximately constant:

$$\Delta \ln c_{t+1} \approx \frac{1}{\sigma(S)} [\ln \beta(S) + r_{t+1}]$$

The net effect on the *level* of consumption is unambiguously downward: higher S means lower c and higher savings.

B Extended Proofs

B.1 Proof of Proposition 3: g/c is increasing in S

Let $R(S) \equiv \psi(S)^{1/\sigma(S)}$. Take logs:

$$\ln R = \frac{\ln(\psi_0 S^\eta)}{\sigma(S)} = \frac{\ln \psi_0 + \eta \ln S}{\sigma(S)}$$

Differentiating with the quotient rule, using $\sigma'(S) = \bar{\sigma} - \underline{\sigma} \equiv \Delta\sigma$:

$$\frac{R'}{R} = \frac{1}{\sigma} \left[\frac{\eta}{S} - \frac{\ln \psi}{\sigma} \Delta\sigma \right]$$

The first term $\eta/(S\sigma)$ is always positive. The second involves $\ln \psi = \ln \psi_0 + \eta \ln S$. For S small enough that $\psi < 1$, $\ln \psi < 0$ and the second term is also positive. For $\psi > 1$ (high S), the second term is negative but of order $\ln S/\sigma^2$, dominated by $\eta/(S\sigma)$ for moderate S . $\square$

B.2 Proof of Proposition 5: Existence of threshold S^F

Define $\Phi(S) \equiv \beta(S)V^{\text{peace}} + \psi(S)(\Omega + \Gamma)$. Then:

- $\Phi(0) = \underline{\beta} V^{\text{peace}}$
- $\Phi(1) = \bar{\beta} V^{\text{peace}} + \psi_0(\Omega + \Gamma)$
- $\Phi'(S) = (\bar{\beta} - \underline{\beta})V^{\text{peace}} + \psi_0\eta S^{\eta-1}(\Omega + \Gamma) > 0$

Φ is strictly increasing and continuous. If $\Phi(0) < R_0$ (revenge is tempting when soulless) and $\Phi(1) > R_0$ (forgiveness dominates when fully Christ-centered), by the intermediate value theorem there exists a unique $S^F \in (0, 1)$ where $\Phi(S^F) = R_0$.

For $S > S^F$: forgiveness is a best response to forgiveness. We also need forgiveness to be a best response to revenge: $\pi^{FR}(S) > \pi^{RR}$, i.e., $-L + \beta(S)V^{\text{recovery}} > \pi^{RR}$. Since $\pi^{RR} < 0$ and $\beta(S)V^{\text{recovery}}$ grows with S , this holds for S sufficiently high.

For $S < S^F$: revenge is a best response to forgiveness. By symmetry, (R, R) is the unique equilibrium. $\square$

Remark. Partial spiritual transformation is dangerous: if only one group crosses S^F while the other remains below, the forgiving group gets exploited. This is why community-wide spiritual renewal is necessary; individual virtue without collective transformation leads to victimization.

B.3 Proof of Proposition 7: Multiple equilibria

With social interaction, the steady-state condition is:

$$(\delta^S - \xi)\bar{S}^* = \phi^S(i^{S*}(\bar{S}^*)) + \bar{G}$$

Define $L(\bar{S}) = (\delta^S - \xi)\bar{S}$ (linear) and $R(\bar{S}) = \phi^S(i^{S*}(\bar{S})) + \bar{G}$ (convex). At $\bar{S} = 0$: $L(0) = 0$ and $R(0) = \bar{G} > 0$, so $R > L$ at the origin.

R is convex because: (1) higher $\bar{S}$ raises μ^S through all four channels and the social multiplier; (2) higher μ^S raises i^{S*} ; (3) the social multiplier reinforces, making the relationship convex.

If $\delta^S - \xi > 0$ (stability condition), L is a line with positive slope. The convex R starts above L , dips below (first crossing = low equilibrium $\bar{S}_L$), curves back up (second crossing = unstable threshold $\bar{S}^*$), and stays above (third crossing = high equilibrium $\bar{S}_H$).

Stability at a crossing requires $L'(\bar{S}) > R'(\bar{S})$. This holds at $\bar{S}_L$ and $\bar{S}_H$ but not at $\bar{S}^*$. $\square$